RED SEAL EXAM PREP

308A Sheet Metal Worker

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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Mock Exam — answer all questions, then check the key

Q1. **When developing a pattern for a cylindrical elbow using parallel line development, what determines the number of equal divisions the circumference is broken into?**

A) The number of gores or pieces making up the elbow

B) The gauge of the sheet being formed and its bend allowance

C) The centreline radius of the elbow divided by the duct diameter

D) The accuracy required in the finished pattern curve

Q2. **A sheet metal worker discovers asbestos-containing pipe insulation adjacent to ductwork being replaced. What is the correct action?**

A) Wear a dust mask and continue with the work

B) Remove the insulation yourself before installing duct

C) Stop work, leave it undisturbed, and notify the supervisor

D) Encapsulate it with duct sealant and proceed

Q3. **A drawing calls for 24 gauge material. One fitting is being cut from galvanized steel and an identical one from 304 stainless. A worker sets a micrometer on each sheet. What should the two readings be?**

A) About 0.70 mm on both, as the gauge number is the thickness

B) About 0.64 mm on both, once the zinc coating is deducted

C) About 0.70 mm on the galvanized and 0.79 mm on the stainless

D) About 0.70 mm on the galvanized and 0.64 mm on the stainless

Q4. **A sheet metal worker is TIG welding a 16-gauge (1.6 mm) 304 stainless steel duct section. The band around the finished weld comes out dark blue and grey. What does this indicate and what is the cause?**

A) Mill scale, so the sheet is carbon steel and not 304

B) Heavy heat tint, from lost shielding gas coverage

C) Ordinary weld colouring on stainless, needing no action

D) The filler rod composition is wrong for 304 stainless

Q5. **In British Columbia, a sheet metal worker is cutting on a layout table when a sheet slips and lacerates his forearm. He is treated at a walk-in clinic, where the wound is closed with sutures, and he returns to finish his shift. Under the Workers Compensation Act, the employer must:**

A) Report the injury; no investigation is required

B) Report the injury and investigate the incident

C) Investigate it, but the clinic reports the injury

D) Record it as first aid, since no time was lost

Q6. **A sheet metal worker is applying flashing near an unguarded roof edge. How does a travel restraint system differ from a fall arrest system?**

A) Restraint prevents reaching the edge; arrest stops a fall

B) Restraint suits sloped roofs; arrest suits only flat ones

C) Restraint needs no anchor; arrest needs a rated anchor

D) Restraint requires a full harness; arrest allows a belt

Q7. **A length of round duct is cut off on a slant at one end. Seen square on, that cut opening is an ellipse. The piece is now developed by the parallel line method: the stretch-out is stepped off in equal divisions, and the height of the cut at each division is projected across from the elevation. What shape do the plotted heights trace along the cut edge of the pattern?**

A) A sine wave, one full cycle across the stretch-out

B) A straight line, since the cut is made on a flat plane

C) A circular arc struck from a single centre point

D) An ellipse, matching the shape of the cut opening

Q8. **A panel is being run through a Pittsburgh lock former to form the pocket on one edge. How is the panel presented to the machine?**

A) Lying flat, the edge fed squarely into the rolls against the guide

B) Standing on edge in the rolls; panel orientation makes no difference

C) Entered from the outfeed side so the pocket runs toward the operator

D) Lying flat, but entered at about 45 degrees to the line of the rolls

Q9. **SMACNA defines three duct sealing classes. What does Sealing Class A require?**

A) Sealing applied to the duct exterior only for condensation control

B) Only transverse joints between duct sections must be sealed

C) All joints, seams and duct wall penetrations sealed

D) No sealant — Class A uses mechanical connections only

Q10. **A project specification adopts the SMACNA HVAC Duct Construction Standards, and a fabricator is choosing fasteners to secure reinforcement angles to galvanized steel duct. Which practice agrees with what SMACNA says about fastenings on steel duct?**

A) Screws may replace specified welds if strength is equal

B) Self-drilling screws are not accepted for duct fastening

C) Stainless fasteners are required on galvanized steel duct

D) Fastenings may not project more than 13 mm into the duct

Q11. **What is a Pittsburgh lock seam and for what application is it primarily used?**

A) A spiral seam used on round ductwork formed by a spiral pipe machine

B) A cross-seam used to connect two sections of rectangular duct end-to-end

C) A double-folded flat seam used on architectural metal roofing

D) A longitudinal seam used to close the sides of rectangular duct sections

Q12. **Before starting any work on an energised electrical panel in the mechanical room where ductwork is being installed, the sheet metal worker must:**

A) Apply LOTO procedures on all energy sources

B) Wear rubber insulating gloves only

C) Simply turn off the panel breaker and proceed

D) Notify the foreman verbally and proceed

Q13. **A sheet metal contractor is running the exhaust duct from laundry-drying equipment in a multi-storey building. Under Article 6.3.2.10. of Division B of the National Building Code of Canada 2020, what does the Code require of that duct?**

A) Independent of other exhaust ducts, cleanable, smooth-walled

B) Tied into the washroom exhaust at the inlet of a common fan

C) Run in flexible corrugated duct, discharging to a service shaft

D) Insulated and discharged into a ventilated attic or roof space

Q14. **An office building draws its return air back through the concealed ceiling space instead of through ducted returns. Under Division B of the National Building Code of Canada 2020, which statement about that arrangement is correct?**

A) A ceiling space must not serve as a return-air plenum

B) A return-air opening must sit at least 2 m above floor

C) A return-air plenum must be lined on all its surfaces

D) A public corridor must not serve as a return-air plenum

Q15. **A rooftop exhaust fan is shut down overnight and outdoor air blows back down the duct into the building. Which unpowered damper is fitted in the duct to stop this?**

A) A manual volume control damper

B) A motorised isolation damper

C) A fire damper with a fusible link

D) A backdraft (gravity) damper

Q16. **Rectangular duct in 22 gauge is being joined with SMACNA transverse joint T-21, the welded flange, on a project whose specification adopts the SMACNA standards. Hangers are not being provided at every transverse joint. What do the notes for that joint then require at the duct corners?**

A) A 6.3 by 102 mm rod brazed or welded in the corners

B) A 16 gauge corner piece closed with 9.5 mm bolts

C) A continuous fillet weld run around every corner joint

D) A gasket lapped at the corner and bolted through

Q17. **ASHRAE 62.1 governs ventilation for acceptable indoor air quality. Which parameter does it specify for commercial occupancies?**

A) Minimum refrigerant charge weight per equipment room

B) Maximum permitted duct velocity in metres per second

C) Minimum outdoor air rate per person and floor area

D) Maximum allowable CO2 concentration in the flue gas

Q18. **When developing the pattern for a right cone using radial line development, what measurement represents the radius of the arc drawn in the pattern?**

A) The diameter of the cone base

B) The slant height of the cone

C) The radius of the cone base

D) The vertical height of the cone

Q19. **According to SMACNA, pressure class for HVAC ductwork is defined by the maximum static pressure the duct system will operate under. Which of the following correctly lists SMACNA low-pressure classes?**

A) Class 1/2, Class 1, Class 2 (in inches of water gauge, WG)

B) Class 100, 200, 300 based on CFM airflow

C) Pressure class is not defined by SMACNA — it is determined by the HVAC engineer only

D) Class A, B, C based on material gauge only

Q20. **A rooftop outdoor air intake is being located for a commercial air handling unit. The nearest source of contaminants is the kitchen cooking exhaust outlet of the restaurant below. Under Table 6.3.2.9. of Division B of the National Building Code of Canada 2020, what is the minimum distance required between them?**

A) 3.0 m

B) 1.5 m

C) 3.5 m

D) 7.6 m

Q21. **An outdoor air intake hood and an exhaust outlet are being fabricated for a rooftop unit. Under Article 6.3.2.9. of Division B of the National Building Code of Canada 2020, the corrosion-resistant screen fitted at each of those exterior openings must have openings no larger than:**

A) 25 mm

B) 50 mm

C) 15 mm

D) 6 mm

Q22. **A rectangular duct is to be built to the SMACNA rectangular duct reinforcement schedule. Which factors together fix the minimum galvanized steel gauge of the duct wall?**

A) Air velocity and the overall length of the duct run

B) Pressure class and duct size, taken by themselves

C) Duct size by itself, whatever the pressure class

D) Pressure class, size and reinforcement spacing

Q23. **Hanger anchors are being shot into a concrete slab with a powder actuated tool. The operator presses the tool to the slab, pulls the trigger, and nothing fires. Under the British Columbia Occupational Health and Safety Regulation, what must be done next?**

A) Lift it clear, aim it downward and open the breech at once

B) Wait the 5 seconds holding it clear of the work, then reload

C) Keep it pressed on the work at least 5 seconds, then unload

D) Press it down and squeeze the trigger again straight away

Q24. **Which welding process is most commonly used for joining stainless steel sheet metal in HVAC and food service applications?**

A) Flux-Core Arc Welding (FCAW) — self-shielding gives the cleanest stainless bead

B) Oxy-acetylene welding — lowest heat input, so stainless never sensitizes

C) Shielded Metal Arc Welding (SMAW/stick) — low spatter and best control on thin gauge

D) Gas Tungsten Arc Welding (GTAW/TIG) — clean, precise welds on thin material

Q25. **A long seam welded as one continuous pass in 20 ga sheet has left the panel buckled between the tacks. Which change best controls the distortion?**

A) Weld short stitches along the seam, backed by chill bars

B) Open the root gap so the joint has more room to move as it heats

C) Run one continuous pass at a lower travel speed to even the heat

D) Preheat the whole panel before welding so it expands evenly

Q26. **A sheet metal crew is sealing a round duct system built from factory-made spiral pipe to a seal class that calls for longitudinal seams to be sealed. Under SMACNA's HVAC Duct Construction Standards (4th edition), what applies to the spiral lockseams of that pipe?**

A) They need a bead of sealant along the outside of the pipe

B) They count as transverse joints and are sealed as joints

C) They must be sealed like every other longitudinal seam

D) They are exempt from sealant; the joints are still sealed

Q27. **A fire damper where a duct passes through a fire separation is arranged to close when a fusible link operates. Under the National Building Code of Canada 2020, Division B, Sentence 3.1.8.10.(3), what temperature rating must that heat-actuated device have?**

A) About 30°C above the system's maximum temperature, running or shut down

B) About 30°C above the average duct air temperature while the fan is running

C) About 55°C above the design temperature of the room housing the damper

D) About 30°C below the temperature at which the duct sheet metal would fail

Q28. **A shop is making slip-and-drive transverse joints on rectangular duct that has to be sealed to a specified sealing class. SMACNA warns that two things about the connection must be controlled or the sealing will be compromised. What are they?**

A) Rivet spacing and the size of the rivet head used

B) Sheet gauge and the pressure class of the section

C) Sealant brand and the temperature at application

D) Connector length and notch depth at the duct ends

Q29. **When working in a ceiling plenum space above a suspended ceiling, what is a key safety concern specific to sheet metal workers?**

A) Energised conduit contact and falls through the grid

B) UV exposure from the fluorescent and LED fixtures overhead

C) Sustained noise above 85 dBA from the operating HVAC equipment

D) Chemical burns from duct sealant curing in the enclosed space

Q30. **Two folded edges on a round duct have been hooked together to form a grooved seam. Which hand tool locks that seam?**

A) A setting hammer worked along the seam

B) A hand seamer, which folds a flat edge

C) A hand groover sized to the seam width

D) A hand crimper worked along the seam

Q31. **What is the "seam line" on a developed pattern?**

A) The reference line from which all element lengths are measured

B) The outer boundary of the developed flat blank, followed when cutting

C) The line the blank is folded on to form a corner of the fitting

D) The line where the pattern is joined to form the fitting

Q32. **When a fitting has both curved and flat surfaces (e.g., a round-end rectangular plenum takeoff), what development approach is used?**

A) Radial line development for every surface

B) The curved section is approximated as flat — no special method needed

C) Triangulation for all surfaces regardless of shape

D) Parallel line for flat sides, triangulation for the transition

Q33. **In SMACNA's notes for the transverse joints of Figure 1-4, one of these joints is held to a 2 in. w.g. maximum static pressure. Which one?**

A) Flat S slip

B) Standing S slip

C) Welded flange

D) Companion angle flange

Q34. **What is the correct method to carry a full sheet of heavy-gauge sheet metal to avoid back injury?**

A) Carry vertically with two workers, close to the body

B) Carry horizontally with arms extended in front of the body

C) Carry flat overhead to keep the centre of gravity high

D) One worker grips the middle of the sheet with both hands

Q35. **A duct fitter is closing a fold by hand on 24 gauge galvanized sheet. Why is a rawhide, hide or plastic mallet used on the sheet rather than a steel hammer?**

A) A steel face stretches metal and cuts the coating

B) A steel face would spark near the duct sealant

C) Steel hammers are barred from coated sheet work

D) A mallet is heavier, so fewer blows are needed

Q36. **What does "beading" do to a flat sheet metal panel?**

A) Adds stiffening ridges that prevent flutter

B) Creates a decorative surface finish only

C) Removes the oxide layer before welding

D) Creates a sealed hemmed edge on the panel

Q37. **A branch is to be cut into an existing 450 mm diameter round main and connected with a saddle fitting. Under SMACNA's round duct specification S3.4 in the HVAC Duct Construction Standards, 2nd edition (1995), what governs that connection?**

A) Branch up to two thirds of the main, with nothing projecting in

B) Any branch size, once the saddle is screwed and then sealed

C) Branch up to half the main, the collar finishing flush inside

D) Branch up to three quarters of the main, a short stub left inside

Q38. **A sheet metal worker is spot welding two 20ga mild steel panels. What parameter most directly controls nugget size?**

A) Electrode material (copper alloy vs carbon)

B) Number of spots per metre

C) Weld current (amperage)

D) Panel orientation (horizontal vs vertical)

Q39. **A 450 mm by 300 mm rectangular duct is fabricated from 24ga galvanised steel in sections, and two sections must now be joined end to end. Which connection does that job?**

A) A Pittsburgh lock run along a duct corner

B) A grooved seam run the length of the section

C) A button punch snap lock at a duct corner

D) A slip-and-drive (S & drive) connection

Q40. **A rectangular duct section 400 mm by 250 mm is to be made from one sheet, closed with a Pittsburgh lock at one corner. What width of flat blank does the layout call for?**

A) 1300 mm, the girth already covering the lock

B) 1300 mm, with the lock allowance taken out of it

C) 1300 mm, plus the pocket and edge allowances

D) 650 mm, plus the pocket and edge allowances

Q41. **What is the squaring arm on a squaring shear used for?**

A) Setting the running clearance between upper and lower blades

B) Holding the sheet square to the blade before the cut

C) Holding the sheet down on the table while the blade cuts

D) Setting the distance the sheet is pushed past the blade

Q42. **On a true-length diagram for a triangulated transition, an element line has a plan (horizontal) length of 400 mm and a vertical height (rise) of 300 mm. What is its true length?**

A) 250 mm

B) 350 mm

C) 700 mm

D) 500 mm

Q43. **A VAV (Variable Air Volume) system modulates airflow to individual zones. What controls the airflow at each zone terminal?**

A) Manual balancing dampers set at commissioning

B) A VAV box with a thermostat-controlled damper

C) A fixed orifice plate in the branch duct

D) The AHU supply fan speed alone

Q44. **What is the purpose of a turning vane installed inside a 90° rectangular duct elbow?**

A) To split the airflow evenly between two branches taking off at the elbow, balancing the system

B) To stiffen the elbow corners so a lighter gauge of sheet can be used at the bend

C) To guide airflow smoothly around the bend, reducing turbulence and noise

D) To break the airstream into layers so condensation cannot form on the elbow's inner wall

Q45. **Which shop machine forms the narrow flanged "male" edge that enters the pocket of a Pittsburgh lock?**

A) Bar folder

B) Slip roll former

C) Pittsburgh lock former

D) Hand notcher

Q46. **When soldering a flat lock seam (folded seam) on tinplate, what is the correct heating technique to ensure full solder penetration?**

A) Heat the seam itself until it melts solder on contact

B) Apply solder to the iron tip, then wipe across the cold seam

C) Melt bulk solder over the seam with a torch and let it flow

D) Heat the solder wire directly until it melts into the seam

Q47. **A 205 L steel drum that previously held a solvent-based duct sealant is to have a lifting bracket welded to its side. Under the BC Occupational Health and Safety Regulation, what has to happen before the arc is struck?**

A) Airing the drum open for a full day before the work

B) Opening the bung and keeping an extinguisher at hand

C) Thorough cleaning, then testing by a qualified person

D) Purging the drum with oxygen and leaving it vented

Q48. **A sheet metal worker is developing a pattern for an offset transition fitting — a rectangular duct offset to one side connecting to a round duct. Which sequence correctly describes the triangulation method?**

A) Draw plan and elevation → divide surfaces into triangles → find true lengths → build the pattern

B) Divide only the round end into segments → project them directly to the square end → connect points

C) Find the apex of the cone → draw radial lines from the apex → swing arcs for each section

D) Draw the elevation only → project parallel lines to a stretch-out line → mark off the perimeter

Q49. **What is the primary purpose of insulating HVAC supply ductwork in an unconditioned attic?**

A) To reduce duct velocity noise in the attic

B) To prevent condensation and reduce thermal loss

C) To meet SMACNA pressure class requirements

D) To allow higher static pressure ratings

Q50. **A duct system shows higher than design static pressure drop across a filter bank. What is the most likely cause?**

A) Duct leakage downstream of the filter

B) Supply fan speed set too low

C) Excessive outdoor air intake volume

D) Filters loaded with dust and dirt

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 120-question 308A bank; the last page shows what the complete printable bank contains.

Q1. Answer: D

In parallel line development the plan view of the duct is divided into equal parts and those same divisions are then stepped off along the stretch-out line, so the circumference divisions and the stretch-out divisions are one and the same set and neither can determine the other. What sets the number is how closely the pattern has to follow the true curve. Each division produces one point where the mitre line is projected across onto the stretch-out, and the pattern outline is drawn by joining those points, so a coarse division cuts corners off the curve while a fine one traces it closely at the cost of layout time. Twelve or sixteen equal parts is the usual shop compromise. The Red Seal Occupational Standard states the criterion outright in sub-task B-6.02: the plan and elevation are divided into equal parts to achieve required accuracy. The number of gores decides how many separate patterns an elbow needs, not how finely each one is divided. Sheet gauge and bend allowance change the flat blank size and the seam and edge allowances, not the spacing of the division lines. Centreline radius divided by diameter is the elbow throat ratio, a fitting-selection figure with nothing to do with layout spacing.

Q2. Answer: C

Asbestos disturbance requires a certified abatement contractor. The worker must stop, leave the area, secure the zone, and notify the supervisor. Continuing work violates OH&S asbestos regulations.

Q3. Answer: D

A gauge number is not a thickness. It is a position in a series, and each material has its own series, so the same number gives a different thickness in each one. In the SMACNA appendix tables, 24 gauge galvanized sheet is 0.0276 in. (0.70 mm) nominal, while 24 gauge stainless is 0.0250 in. (0.635 mm), and 24 gauge uncoated steel on the manufacturers' standard gauge is 0.0239 in. (0.607 mm). Three different thicknesses, one gauge number. That is exactly why the sheet is measured with a micrometer and why the tables give a decimal thickness beside every gauge number, the number itself being carried along only because the trade is used to it. The tempting wrong answer is that the difference is the zinc: it is not. A G60 coating is in the order of 0.001 in. (about 0.025 mm) over both faces together, while the gap between the two sheets here is about 0.065 mm, more than double that, and the stainless has no coating to deduct in the first place. Stainless being a heavier, stronger material does not make its gauge series thicker either; its series runs thinner than the galvanized one at the same number. One more thing the micrometer settles: gauge thickness carries a tolerance band, 24 gauge galvanized running from about 0.60 mm to 0.80 mm, and the duct construction ratings are worked from the minimum thickness rather than the nominal, so a reading near the bottom of the band is what has to satisfy the job.

Q4. Answer: B

Heat tint is chromium oxide, formed when hot stainless is exposed to oxygen, and its colour is a rough measure of how much oxide grew and how much chromium was drawn out of the surface underneath to build it. A straw or light gold tint is minor and is normally accepted on non-critical HVAC work. Dark blue and grey mean heavy oxidation, and the chromium-depleted layer beneath that tint no longer carries the corrosion resistance of the parent metal, so at this colour the tint has to be taken off and the shielding put right. The cause is oxygen reaching the metal while it is still hot: too little argon flow, too great a distance from cup to work, a torch angle that draws air into the shielding stream, or moving the torch away before the post-flow has cooled the weld under gas. Mill scale is a different thing altogether. It is the oxide skin a hot rolling mill leaves over a whole sheet before any welding is done, dull and uniform across the surface, whereas what is described here appeared during the weld and is graded, running from straw at the outer edge through gold and purple to blue nearest the fusion line. Stainless duct stock is supplied cold rolled and pickled with a bright finish, so there is no rolling scale on it to find. The filler rod is not the cause either: composition governs how the deposited weld metal behaves and shows up as cracking or as poor corrosion behaviour in the bead itself, not as a graded colour band on the parent metal beside it. Prevention: correct flow rate, a gas lens, correct torch angle, and enough post-flow to cool the weld under shielding, with the root side back-purged where it must stay bright.

Q5. Answer: B

What decides this is medical treatment, not lost time. A nick dressed at the shop and entered in the first aid record goes no further, but a wound closed with sutures at a clinic is health care, and it engages two separate employer duties that run alongside each other. On reporting: section 150(1) of the Workers Compensation Act requires an employer to report to the Board, WorkSafeBC, within three days after its occurrence, every injury to a worker that is or is claimed to be an injury arising out of and in the course of the worker's employment. That duty does not wait to see whether a shift is lost, and it is the employer's own - whatever report a clinic or an attending physician makes on its own account, section 150(6) makes an employer who fails to report guilty of an offence. Section 150(7)(a) allows the Board to except a category of minor injuries from reporting by regulation, which is the space a first aid entry occupies, but the Act sets the investigation threshold at injury requiring medical treatment, and closing a wound with sutures is medical treatment. On investigating: section 69(1)(b) puts an incident that resulted in injury to a worker requiring medical treatment among those the employer must investigate. Section 71 requires the preliminary investigation immediately, with its report completed within 48 hours, and section 72 the full investigation, whose report goes to the Board within 30 days. Nothing in that turns on the injured worker asking for it. Other provinces impose their own versions of both duties, on their own bodies and their own time limits, which is why the scenario names a province.

Q6. Answer: A

The two systems answer different questions. A travel restraint system is rigged so the worker physically cannot get to the edge: the lanyard is short enough that the fall never happens. A fall arrest system assumes the worker will go over and is built to stop the fall before they strike anything below. That is why the regulations rank restraint ahead of arrest. The Ontario construction regulation requires a guardrail system first, and where that is not practicable, the highest ranked practicable method from travel restraint, then a fall restricting system, then fall arrest, then a safety net. Roof slope does not decide which system is used. Both must be attached to a fixed support, so a restraint lanyard is anchored just as an arrest lanyard is. The hardware difference runs opposite to the common belief: travel restraint may use a full body harness or a safety belt, while fall arrest requires a full body harness with a shock-absorbing lanyard, because arresting a fall in a belt injures the worker.

Q7. Answer: A

Two different curves are in play and the item turns on keeping them apart. The cut face, the opening seen square on, is a plane section of a cylinder and is an ellipse. The cut edge on the flat pattern is a different curve, because developing the piece lays the circumference out straight while leaving every height as it was. Why it comes out as a wave is visible in the elevation. The mitre line there is straight, and each height is read where that straight line crosses an element. But the elements do not project evenly onto the elevation: they crowd together toward the two outside edges of the pipe and open out through the middle, so a point standing at an angle theta around the pipe projects to a horizontal position proportional to the cosine of theta, and the straight mitre line therefore hands it a height proportional to the cosine of theta as well. On the stretch-out those same points are spaced evenly, one equal division apart. Replotting cosine heights at even spacings is exactly what turns the straight line into a wave: measured from the mid-height of the cut, the height is the radius times the tangent of the cut angle times the cosine of the angle travelled around the pipe, plotted against arc length. That is a sinusoid, and it runs one full cycle over the stretch-out, with a crest at the longest element, the mid-height crossed at the two side elements, and a trough at the shortest element. Whether it reads as a sine or a cosine depends only on which element the stretch-out is started from; the curve is the same. An ellipse answers a different question. It is the true shape of the hole, and it is what the cut looks like while the metal is still round. Setting an ellipse out on the blank instead gives a piece whose end will not lie in a plane once it is rolled up. No circular arc will fit the curve, whatever radius is chosen, because an arc bends the same way from end to end. This curve does not: it arches over the longest element and hollows under the shortest, reversing at the two side elements. That is why the layout is finished by joining the plotted points with a flexible curve rather than by swinging an arc from a centre. A straight line would be right for a prism, not a cylinder. Rectangular duct cut on a slant does develop with straight pattern edges, one straight run across each face, and that is where the expectation comes from. On round duct the mitre line is straight only in the elevation, where the elements are foreshortened; once the even stretch-out spacings replace those foreshortened ones, the line becomes a wave.

Q8. Answer: A

A Pittsburgh lock former does not fold the pocket in one action. The edge passes through a train of roll sets, each bending it a little further than the last, so the pocket is formed progressively along the whole length as the panel travels through. That is why presentation matters. The panel lies flat on the machine table and the edge to be formed is fed straight into the first roll set and kept tight against the guide, so the same amount of metal enters every pass and the finished pocket is the same depth from one end of the panel to the other. Start the edge in at an angle and the amount of metal in the rolls changes as the panel goes through, which tapers the pocket and can jam or spring the roll set. Standing the panel on edge takes the flat off the table and leaves nothing keeping the panel travelling straight, so orientation is very much not arbitrary. And the rolls are driven one way only, from infeed to outfeed, so a panel cannot be entered from the outfeed side at all.

Q9. Answer: C

SMACNA Table 1-2, Standard Duct Sealing Requirements: Class A = all transverse joints, longitudinal seams AND duct wall penetrations, for the 4 in. w.g. construction class and up. Class B = all transverse joints and longitudinal seams only, 3 in. w.g. Class C = transverse joints only, 2 in. w.g. Duct of the 1 in. and 1/2 in. w.g. classes is not required by the manual to be sealed unless the designer calls for it, although variable air volume duct of those classes upstream of the boxes must meet Class C. Helical (spiral) lock seams are exempt from sealant requirements.

Q10. Answer: D

SMACNA's standard specification S1.41 for rectangular duct transverse joints states that fasteners used on steel duct shall be steel, that standard or self-drilling sheet metal screws may be used as appropriate, that blind rivets using pull-through mandrels are not permitted if they leave holes for air leakage, that fastenings shall not project into duct interiors more than one half inch, which is 13 mm, and that where only bolts or welds are specified, other types of fastening are not allowed. The projection limit is the one that decides this question. A screw driven through an angle and a duct wall leaves its point inside the airstream, and a long screw point catches lining, tears filters and gathers lint, so the standard caps how far it may stand in. The near miss is substituting screws for a specified weld on the argument that the strength works out the same. SMACNA closes that door in the same paragraph, because a specification that calls out bolts or welds has usually done so for a reason the fastener catalogue cannot see, such as fire exposure, service temperature, vibration or a leakage class the perforation would break. The other two are misreadings of the same paragraph in the other direction. The standard calls for steel fasteners on steel duct rather than stainless, and it accepts standard and self-drilling sheet metal screws alike. Remember that SMACNA is a construction standard rather than Canadian law; it governs this work because the project specification adopts it, or because the authority having jurisdiction has.

Q11. Answer: D

The Pittsburgh lock (also called Pittsburgh seam) is a longitudinal closing seam for rectangular ducts. SMACNA designates it Type L-1 among its longitudinal seams for rectangular duct - Figure 2-2 in the current 4th edition, Fig. 1-5 in the 1995 2nd edition, whose notes give it for use on straight ducts and fittings. One edge is formed into a pocket (the "Pittsburgh pocket"), and the mating edge, a single flange bent to about a right angle, is inserted and then clinched (hammered) closed, giving a mechanical interlock along the full length of the duct. The 1995 edition also notes that leakage at rectangular longitudinal seams of the Pittsburgh and snaplock types is low compared with that in transverse joints.

Q12. Answer: A

LOTO (Lockout/Tagout) is mandatory before working near or on energised equipment. The sheet metal worker must de-energise, lock out, and verify zero energy state.

Q13. Answer: A

Sentence 6.3.2.10.(7) gives three requirements for an exhaust duct connected to laundry-drying equipment: it shall be independent of other exhaust ducts, accessible for inspection and cleaning, and constructed of a smooth corrosion-resistant material. Sentence (9) adds that such ducts or vents shall discharge directly to the outdoors. Every one of those requirements is about lint. Lint settles wherever the air slows or the wall is rough, so a ribbed or flexible wall packs with it, a duct shared with another system spreads it, a duct nobody can open is never cleaned, and lint at a dryer's operating temperature is the fuel for a duct fire. Tying the dryer into the washroom exhaust is the near miss because that arrangement is genuinely permitted elsewhere in the same Article: Sentence (12) lets exhaust systems serving rooms with water closets, urinals, basins, showers or slop sinks, and those serving rooms with residential cooking equipment, be interconnected at the inlet of an exhaust fan when suitable back pressure devices stop odours crossing over. Laundry-drying equipment is not in that permission. Discharging into an attic or roof space is not a discharge to the outdoors at all and dumps the dryer's moisture into the roof assembly. Where several dryers are vented collectively, Sentence (8) sets out the only route: one common exhaust duct vented by one central exhaust fan, an interlock that starts that fan when the laundry-drying equipment is in use, and makeup air provided to the space.

Q14. Answer: D

The Code deals with this in two short sentences. A ceiling space may be used as a return-air plenum, and where it is, the plenum requirements of Article 3.6.4.3. apply to it: everything inside the space needs a flame-spread rating not more than 25 and a smoke developed classification not more than 50 apart from a short list of exceptions, the ceiling supports must be noncombustible, and where the ceiling membrane forms part of a required fire-resistance rating, every opening through it is protected by a fire stop flap that shuts off airflow into the space in a fire. What the Code prohibits outright is different: a public corridor or an exit shall not be used as a return-air plenum. The reason is that a corridor and an exit are the route people leave by, and drawing return air along them pulls smoke into the very space that has to stay usable. Saying no ceiling space may be a plenum is the near miss, and it is simply too broad; the ceiling plenum return is a normal design, it just comes with conditions. There is no Code requirement to line a return-air plenum, and linings that are used must not interfere with volume, balancing or fire dampers. The 2 m figure belongs to a different rule, which does not ban low openings but requires supply, return and exhaust openings less than 2 m above the floor to be protected by grilles whose openings will not pass a 15 mm diameter sphere.

Q15. Answer: D

A backdraft damper, also called a gravity damper, has light blades hung on a horizontal axis and held at or near the closed position by their own weight, sometimes with a small counterweight to set how easily they lift. The fan's airflow pushes them open in the direction of flow, and when the fan stops they fall shut, so air cannot travel back down the duct. Nothing drives them - no actuator, no wiring, no interlock with the fan - which is the condition the stem sets. A motorised isolation damper will also shut an idle duct, and on a large system it often does, but it needs power and a control signal, so it does not answer a question about an unpowered device. A fire damper sits open in normal service and closes only when its fusible link releases at its rated temperature; it protects a fire separation and takes no notice of which way the air is going. A manual volume control damper is set by hand during balancing and locked in that position, so it neither opens on forward flow nor closes against reverse flow. All four are damper types a sheet metal worker fabricates; what separates them here is how each one is operated.

Q16. Answer: A

The notes for Figure 1-4, joint T-21, welded flange, call for a 5/8 inch, that is 16 mm, minimum flange with an edge weld, and add that on 24, 22 and 20 gauge the fabricator is to braze or weld a 1/4 inch by 4 inch rod, 6.3 by 102 mm, in the corners or else provide hangers at every joint. The reason sits in how this joint carries load. On light gauge there is no angle and no bolted frame; the turned-up flange and its edge weld are the whole joint, and the corner is where two flanges meet at right angles and where the weld tears first when the duct is hung between joints and the section works. A short rod brazed or welded across each corner ties the two flange legs together and puts metal where the tearing starts, and the alternative the note allows is to take the load off the joint entirely by hanging at every one of them. The 16 gauge corner piece with 9.5 mm bolts is the near miss and a genuine SMACNA requirement, but it belongs to a different joint, T-24, the formed flange, where mating flanges are rolled on the duct ends to form a tee and steel corner pieces with 3/8 inch minimum bolts close the corners. Reading a T-24 corner rule onto a T-21 welded flange is the mistake this question is built on. Running a continuous fillet weld around every corner puts a great deal of heat into thin sheet, distorts the flange and is not what the note asks for. A gasket seals a joint and adds no strength at all, and the corner problem here is structural. Note also the related T-21 rule that flanges larger than 16 mm must be spot welded, bolted, riveted or screwed as required to prevent separation and weld breakage.

Q17. Answer: C

ASHRAE 62.1 sets minimum outdoor air ventilation rates by occupancy type, expressed as a rate per person plus a rate per unit of floor area. That is the Ventilation Rate Procedure in Section 6.2: Vbz = Rp x Pz + Ra x Az, where Rp (per person) and Ra (per unit floor area) are read from Table 6.2.2.1 in L/s or CFM. **It carries weight in Canada only because the Code adopts it: Sentence 6.3.1.1.(2) of Division B of the 2024 Ontario Building Code requires outdoor air to be supplied in accordance with Section 6.2 (Ventilation Rate Procedure), Section 6.3 (Indoor Air Quality Procedure) or Section 6.4 (Natural Ventilation Procedure) of ANSI/ASHRAE 62.1, and Sentence 6.3.1.1.(3) requires exhaust ventilation in accordance with Section 6.5.** In the 2012 OBC the same requirement sat at Sentence 6.2.2.1.(2), so older references and study notes will quote that number. Always use the edition your provincial code references, not the newest ASHRAE printing: the 2024 OBC and the National Building Code 2020 reference ASHRAE 62.1-2016, while the 2012 OBC referenced 62.1-2010. Because the 2016 edition dropped dwelling units from its scope, residential ventilation is covered by ASHRAE 62.2 or by Subsection 9.32.3 and CAN/CSA-F326, not by 62.1. The distractors belong to other documents entirely: duct velocity is a design and noise matter, refrigerant charge limits come from ASHRAE 15 and CSA B52, and flue gas CO2 is a combustion analysis reading. None of the three appears in 62.1.

Q18. Answer: B

In radial line development of a cone, all surface lines radiate from the apex. The radius of the arc in the flat pattern equals the slant height — the true length from the apex to the edge of the base circle. The arc length then equals the circumference of the base (2πr). The base radius fixes the included angle of that arc, not its radius, and the vertical height alone is always shorter than the slant height.

Q19. Answer: A

SMACNA classifies duct systems by static pressure: 1/2" WG, 1" WG, 2" WG, 3" WG, 4" WG, 6" WG, and 10" WG. The pressure class determines sheet metal gauge, seam types, reinforcement requirements, and joint spacing. Low-pressure residential systems are typically 1/2" or 1" WG.

Q20. Answer: A

Table 6.3.2.9., Minimum Distances of Air Intakes from Sources of Contaminants, gives a separate figure for each named source, and kitchen cooking exhaust is listed at 3.0 m. A vent for combustion products carries the same 3.0 m. The near miss is 3.5 m, which belongs to a sanitary vent, not to cooking exhaust; the two sit next to each other in the table and are easy to swap. The other figures in the same table are 1.5 m for a driveway, street or parking space, 4.5 m for the entry of a garage for five or more motor vehicles and for a garbage storage or pick-up area, and 7.6 m for a truck loading area or dock and bus parking, for a thoroughfare, arterial road, freeway or highway, and for the discharge of an evaporative heat rejection system. Two further sentences of the same Article govern the same intake. Sentence (3) requires outdoor air intakes to be installed not less than 0.3 m above roofs, landscape grades or other surfaces, taking anticipated snow accumulation into account, so on a Canadian roof the intake often has to stand well above that minimum. Sentence (2) also ties the location back to air quality, requiring the intake to be placed so the air entering the building meets the Code's quality provisions, which is the catch-all when the actual contaminant source is not one of the nine listed.

Q21. Answer: C

Sentence 6.3.2.9.(4) requires exterior openings for outdoor air intakes and exhaust outlets to be shielded from the entry of snow and rain and to be fitted with corrosion-resistant screens of mesh having openings not larger than 15 mm, except where experience has shown that climatic conditions require larger openings to prevent the screen openings from icing over. That exception is written for Canadian winters and is the reason a finer mesh is the wrong answer rather than a safer one: a 6 mm mesh has less free area, blinds off faster with snow and frost, and once it ices the intake is closed. Sentence (5) then requires the screens to be accessible for maintenance, which is what makes the hood's screen removable rather than riveted in place. Do not confuse this with Sentence (1) of the same Article, which uses the same number for a different purpose: supply, return and exhaust air openings located less than 2 m above the floor inside a room are protected by grilles whose openings will not pass a 15 mm diameter sphere. One rule is about weather, birds and debris at the building envelope; the other is about fingers and objects inside occupied rooms.

Q22. Answer: D

The SMACNA rectangular duct reinforcement tables are entered on the table for the pressure class, in the row for the duct dimension — the greater dimension of a duct determines the duct gauge for all four sides, reinforced or unreinforced. Each row then runs across a series of reinforcement spacing options from 10 ft down to 2 ft, and each cell gives a minimum duct gauge as a number and a minimum reinforcement grade as a letter. The manual states that 'any cell within a row is an acceptable selection' for that duct width, so gauge and reinforcement spacing are chosen together: closer reinforcement permits a lighter wall. At 1/2 in. w.g. a 24 in. duct is 22 gauge left unreinforced, or 26 gauge with grade C reinforcement out to 6 ft spacing and grade B at 5 ft and closer. Air velocity and run length drive friction loss and noise, not the wall gauge.

Q23. Answer: C

A load that does not fire may still be about to. A hangfire can go off a moment after the trigger is pulled, so the Regulation requires the operator to hold the tool firmly against the work surface for at least 5 seconds and then follow the manufacturer's instructions for removing the load. Holding the muzzle on the work for that interval means that if the load does fire late, the fastener goes where it was always going to go, into the slab, and the tool stays under control. The near miss is the answer that keeps the 5 second wait but lifts the tool off the work: the wait only does its job while the muzzle is confined, and a live tool held clear is pointed at whatever is beyond it, which is why the same Part of the Regulation says the tool must never be pointed at a person. Opening the breech straight away puts a hand at a cartridge that has not been ruled out, and squeezing the trigger again immediately is not the manufacturer's misfire procedure either; the tool is cleared the way its instructions say, and those instructions and the power load chart have to be on hand at the job in the first place.

Q24. Answer: D

GTAW (TIG welding) is the preferred process for stainless steel sheet metal because it provides precise heat control, produces clean welds without spatter, and maintains the corrosion-resistant properties of stainless steel. SMAW and FCAW create too much spatter and heat distortion on thin gauge material. Oxy-acetylene may cause carbide precipitation in stainless steel.

Q25. Answer: A

Distortion in light gauge is shrinkage: the weld metal and the heat-affected zone contract as they cool and pull the panel out of plane. Two levers control it, put less heat in and take heat out faster. Short stitch welds distributed along the seam let each deposit cool before the next one goes down beside it, so the shrinkage forces are scattered instead of accumulating along one line, and a chill bar or chill plate clamped behind the joint draws heat out of the sheet. A strongback clamped across the joint holds it straight while this happens. The three wrong answers all add heat or add filler. Slowing travel speed raises the heat put into every millimetre of seam. Opening the root gap means more filler metal in the joint and therefore more metal shrinking as it freezes. Preheating a thin panel puts heat into the very part that is already distorting.

Q26. Answer: D

SMACNA exempts helical (spiral) lockseams from the sealant requirement, so a seal class that calls for sealed longitudinal seams still leaves the spiral lockseam of factory-made pipe dry - while the joints in the run are sealed. SMACNA's own presentation of the current 4th edition lists Seal Class B as all transverse joints and all longitudinal seams, required at 3 in. w.g., with spiral seams exempt, and SMACNA's 2021 technical bulletin gives the reason: applying sealant in a spiral seam can result in poor seam closure. The terms come from SMACNA's definition of sealing (S1.9 in the 1995 2nd edition, the edition posted on law.resource.org): a seam joins two longitudinally oriented edges of duct surface material between two joints, helical (spiral) lock seams are exempt from sealant requirements, and all other duct wall connections are joints - girth joints, branch and subbranch intersections, duct collar tap-ins and fitting subsections - which are sealed as the seal class requires. So the spiral lockseam is not a transverse joint, it is not sealed like the other longitudinal seams, and the exemption does not call for a bead along the outside of the pipe instead.

Q27. Answer: A

Sentence 3.1.8.10.(3) requires the fusible link or other heat-actuated device to have a temperature rating **approximately 30°C above the maximum temperature that would exist in the system, whether it is in operation or shut down**, and to be located where it is readily affected by an abnormal rise in temperature in the duct. The reference point is the system's *maximum* temperature under both conditions, so a link sized only to the average air temperature while the fan runs is wrong twice: it ignores peak temperatures and ignores the shut-down condition, when a duct near a heat source can sit hotter than it does with air moving, risking a nuisance closure. The Code does not rate the link against the room the damper sits in, and 55°C is not its margin. Rating the link below a duct failure temperature sounds protective but is not the Code's method; the rule is a margin above the highest normal system temperature, so the damper stays open in normal service and closes on an abnormal rise.

Q28. Answer: D

SMACNA's text on sealing says it directly: contractors using slip and drive connection systems must control connector length and notch depth on rectangular duct ends, and failure to do so will compromise seal effectiveness. The reason is the corners. A drive cut long, or an end notched deeper than it needs to be, leaves an opening at the corner where the S-cleat, the drive and the two duct ends all meet, and that opening is the hardest place on the joint for a sealant to bridge. Liquid sealants are the type used on slip joints because they fill the small space between overlapping metal, but they are 30 to 60 percent volatile solvent and shrink considerably as they dry, so where the metal clearance exceeds 1/16 in (1.6 mm) several applications may be needed just to fill the void left behind. A corner gap opened by careless notching is far wider than that. Sealant brand and application temperature matter to the product's own performance but they are not what SMACNA names here, sheet gauge and pressure class govern the construction rather than the sealing of the corner, and rivet dimensions belong to a fastened lap rather than to a slip-and-drive joint, which is not riveted.

Q29. Answer: A

Plenum spaces contain energised conduit, communication cables, and the ceiling grid — contact with live wiring or stepping through the grid are key hazards requiring careful footing and awareness.

Q30. Answer: C

A grooved seam is made by hooking two folded edges together and then driving the raised hook down so it locks. The tool for that is a hand groover, a hardened steel bar with a groove milled across its face. The groove is set over the raised lock and struck with a mallet: the tool drives the seam down while the groove itself keeps the lock from being crushed flat. Groovers come in sizes and the groove must match the seam width, because too narrow a groove mashes the lock and too wide a groove will not seat on it. The wrong answers are all real shop tools doing other jobs. A hand seamer has flat jaws. It folds an edge and flattens it, and on a Pittsburgh lock it does close the pocket over the inserted edge, but here the lock stands proud of the sheet and flat jaws have no groove to seat over it, so the seamer would crush the raised hook instead of driving it home. A hand crimper reduces the end of a round pipe so it slips inside the next section. A setting hammer struck directly on a raised grooved seam flattens the lock and can split light gauge metal; it is for dressing a seam already driven home, not for locking one.

Q31. Answer: D

The seam line is where the two edges of the flat pattern come together when the blank is rolled or folded up into the fitting, and where the lock, groove or weld is then made. It is normally laid out on the shortest element of the pattern, the throat of an elbow for instance, so that the seam is as short as it can be and the least material and labour go into closing it. The other lines on the same pattern each do something different: the outline of the blank is the cut line, the stretchout or base line is the reference the element heights are measured from, and the bend lines mark where the metal folds. None of those three is the seam.

Q32. Answer: D

Compound fittings use combined methods: parallel line for developable flat/cylindrical faces and triangulation for non-developable transition zones.

Q33. Answer: A

SMACNA's notes for Figure 1-4 give joint T-5, T-6, flat S slips, a line of their own: the slip is 24 gauge minimum and no lighter than two gauges below the duct, and when it is used on all four sides it is fastened within 2 in. of the corners and at 12 in. maximum intervals, at 2 in. w.g. (500 Pa) maximum static pressure. No other joint offered here is held that low. The standing S connectors, joints T-10 through T-12, are worked to 4 in. w.g. and the same notes say they are not recommended above that. The welded flange, joint T-21, is given no pressure ceiling at all; its note deals with flange width, edge welding and corner reinforcement. The companion angle, joint T-22, is plainly meant to go higher, because its bolt schedule sets 5/16 in. minimum bolts at 6 in. maximum spacing on the 4 in. w.g. class or lower and 4 in. maximum spacing on higher pressures, and a joint would not need a rule for higher pressures if it were never used there. Read all of this as selection limits from a construction standard rather than as Canadian law: SMACNA governs where a project specification or a provincial code adopts it.

Q34. Answer: A

Sheet metal should be carried vertically with two workers — one on each end — for heavy sheets, keeping the load close to the body and using proper lift technique (back straight, knees bent) to prevent cuts and back strain.

Q35. Answer: A

A mallet with a rawhide, hide, plastic or hardwood face delivers the blow without a hard, narrow steel face driving into the metal. On light gauge sheet a steel hammer face does two things at once: it thins and stretches the metal locally, so the panel is left dished and the fold line wanders, and it dents through the zinc, leaving a bright break in the coating that the sealant or paint then has to cover. A mallet spreads the blow, closes the fold and leaves the surface flat. That is why the Red Seal tool list for this trade carries mallets alongside raising, setting, ball peen and riveting hammers. The steel hammers have their own work - driving another tool such as a hand groover, setting rivets against a dolly, dressing a seam that is already formed - but they are not swung onto an open panel face. A mallet is generally lighter than a steel hammer of the same head size, not heavier, so fewer blows is not the reason. Sparking is not the issue either: duct sealants are applied cold and mallets are not chosen here as non-sparking tools. And no rule bars steel hammers from work on coated sheet; they are in constant use on it, driving groovers and setting rivets, just not swung onto an open panel face.

Q36. Answer: A

Beading rolls a small ridge into the panel surface, greatly increasing stiffness by the same principle as corrugation. It prevents oil-canning (panel flutter) in HVAC ductwork. These linear indentations also stiffen the panel against buckling.

Q37. Answer: A

S3.4 of SMACNA's HVAC Duct Construction Standards, 2nd edition (1995), accepts both the saddle and the direct connection of a branch into a larger duct, then puts conditions on them. The diameter of the branch shall not exceed two thirds of the diameter of the main, so on this 450 mm main the branch may go to 300 mm and no further. Protrusions into the interior of the main are not allowed, so no stub of collar may be left standing inside the main. A direct connection of a branch into a main shall include mechanical attachment sufficient to maintain the integrity of the assembly. And all saddle fittings shall be sealed at all pressures - the clause makes no pressure exception. Reading the wrong answers against S3.4: three quarters of the main is over the two-thirds limit, and a short stub inside is a protrusion into the main; half the main is a tighter limit than the clause sets; and screwing and sealing a saddle does not lift the two-thirds limit on the branch diameter, which the clause states without exception. The clause is quoted here from the 1995 edition; check it against the edition your project specification names.

Q38. Answer: C

Spot weld nugget size is primarily controlled by current (amperage). Higher current → larger nugget. Weld time and electrode force also affect nugget; current is the dominant variable.

Q39. Answer: D

Duct connections come in two families and the exam expects them kept apart. A longitudinal seam runs along the length of a section and closes the flat sheet into its rectangular shape at a corner. The Pittsburgh lock, the button punch snap lock and the grooved seam are all longitudinal seams, made in the shop before the section ever leaves it, and none of them joins one section to another. A transverse joint runs around the duct at the end of a section, and that is the connection made end to end. On low-pressure rectangular duct the common transverse joint is slip and drive: an S-cleat takes one duct edge in each of its two channels, and a drive cleat, a single channel, is hammered over both mating edges on the remaining sides with the tabs at its ends bent over the corners. The Red Seal Occupational Standard keeps the two families apart in its own wording, listing forming longitudinal seams and forming transverse seams as separate performance criteria under fabricating ductwork, and installing or forming transverse joints again under assembly. Note also that a pocket lock is a transverse joint while a snap lock is a longitudinal seam; the two names are often run together and they are not the same connection.

Q40. Answer: C

The stretch-out, or girth, of a rectangular duct is the distance right around its cross-section: twice the width plus twice the depth, so 2 x (400 + 250) = 1300 mm. That figure covers the finished surface of the duct and nothing else. The Pittsburgh lock is made from metal that is not part of that finished perimeter. The pocket rolled on one edge and the plain folded edge that enters it are both extra metal, so their allowances are added to the girth rather than found inside it, and the same is true of any hem or flange added at the ends. Taking the lock allowance out of the girth runs the calculation backwards and leaves the section short by the depth of the lock, which cannot be made up once the sheet is cut. Halving the girth to 650 mm would be right only if the section were made from two L-shaped halves seamed at two corners, which is a different construction from the one the question describes. The Red Seal Occupational Standard puts the rule in the performance criteria for pattern development: the pattern points are connected, and allowances for seams and edges are added.

Q41. Answer: B

The squaring arm is the graduated arm that projects from the front of a squaring shear at ninety degrees to the blade. The sheet is pushed up against it so the edge being fed is square to the cut, which is what makes successive cuts on a blank true to one another, and the scale along the arm then lets the operator gauge the width of a piece taken from the front without measuring each one. It positions the work; it does not clamp it. Clamping is the hold-down, a bar or set of feet that comes down on the sheet just ahead of the blade an instant before the cut so the sheet cannot creep or lift. Running clearance between the upper and lower blades is a machine setting made at the blade mounts according to material thickness, and it is what decides whether the cut edge comes off clean, rolled over or chipped. Setting how far the sheet travels past the blade is the job of the back gauge, the stop behind the blade. That is a different device from the squaring arm, and it is the same device a press brake carries to repeat a flange width.

Q42. Answer: D

True length is the hypotenuse of a right triangle whose base is the plan length and whose height is the vertical rise: TL = $\sqrt{(400^2 + 300^2)}$ = $\sqrt{(160000 + 90000)}$ = $\sqrt{250000}$ = 500 mm. This 3-4-5 relationship is the basis of the true-length diagram used in triangulation.

Q43. Answer: B

VAV boxes contain a motorised modulating damper controlled by the zone thermostat. They vary airflow from maximum to minimum (or zero) to maintain setpoint temperature.

Q44. Answer: C

A square elbow without turning vanes creates significant turbulence, high pressure drop, and noise as the airflow separates from the duct walls at the sharp bend. Turning vanes (sheet metal blades inside the elbow) guide airflow in a smooth arc, greatly reducing separation, pressure drop, and noise. Required by many specifications for critical systems.

Q45. Answer: A

A bar folder bends the edge of a sheet to a set width and angle in one stroke. The wing is adjustable, so the same machine will make a light break, a square flange or a flat hem, and because the width is set on the machine every panel comes off with an identical edge. The male half of a Pittsburgh lock is exactly that kind of edge: a single flange bent to about a right angle, which is entered into the pocket before the lip of the pocket is closed over it. The Pittsburgh lock former makes the other half of the same seam, the pocket, not the plain flange that goes into it. A slip roll former curls flat sheet into cylinders and cones between three rolls and puts no sharp fold in the metal at all, which is the whole point of it. A notcher only removes metal at the corners so that folded edges can meet without bunching; it forms nothing.

Q46. Answer: A

Correct soldering: heat the base metal (the seam), not the solder. When the seam is hot enough, touch the solder wire to the metal — it melts and wicks into the joint by capillary action.

Q47. Answer: C

Two subsections apply and both must be satisfied. BC OHS Regulation 12.116(1) states that a container which may have held a combustible substance must be thoroughly cleaned before any welding or burning operation is carried out on it. Subsection (2) then states that burning, welding or other hot work must not be done on any vessel, tank, pipe or structure, or in any place where the presence of a flammable or explosive substance is likely, until tests have been made by a qualified person to ensure the work may be safely performed and suitable safe work procedures have been adopted, including additional tests at intervals that keep the workers safe. Cleaning alone is therefore not enough; the atmosphere inside has to be proven, and proven again as the work goes on, because a warming drum keeps driving vapour out of residue that cleaning missed. Airing the drum open for a day is the near miss, and it is the practice that kills people. Solvent soaks into seam laps and sits under the rolled chime where nothing rinses it, so a drum can read clean, smell clean and still hold a vapour concentration in the flammable range; time passing proves nothing, and only a test by a qualified person does. Purging with oxygen is the dangerous inversion of a real technique. The recognised purges are an inert gas or filling the drum with water to drive the vapour space out; oxygen does the opposite, enriching the atmosphere so that anything ignitable burns far more violently. Keeping an extinguisher within reach is required at any welding location by section 12.126, but an extinguisher is a response after ignition, and a drum that lets go is an explosion rather than a fire to be put out.

Q48. Answer: A

Triangulation sequence: 1) Draw plan and elevation views. 2) Divide the surface into triangles (diagonals across each face). 3) Find the true length of each line using a true length diagram (rotate each line to horizontal and measure). 4) Draw each triangle in sequence using the true lengths, building the flat pattern triangle by triangle.

Q49. Answer: B

Insulation on supply ducts in unconditioned spaces prevents condensation (cold ducts in humid attics), reduces heat gain from the attic, and improves system efficiency.

Q50. Answer: D

A clogged/loaded filter has high resistance, causing elevated pressure drop across the filter section. This also reduces airflow. Filters must be replaced or cleaned.

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